

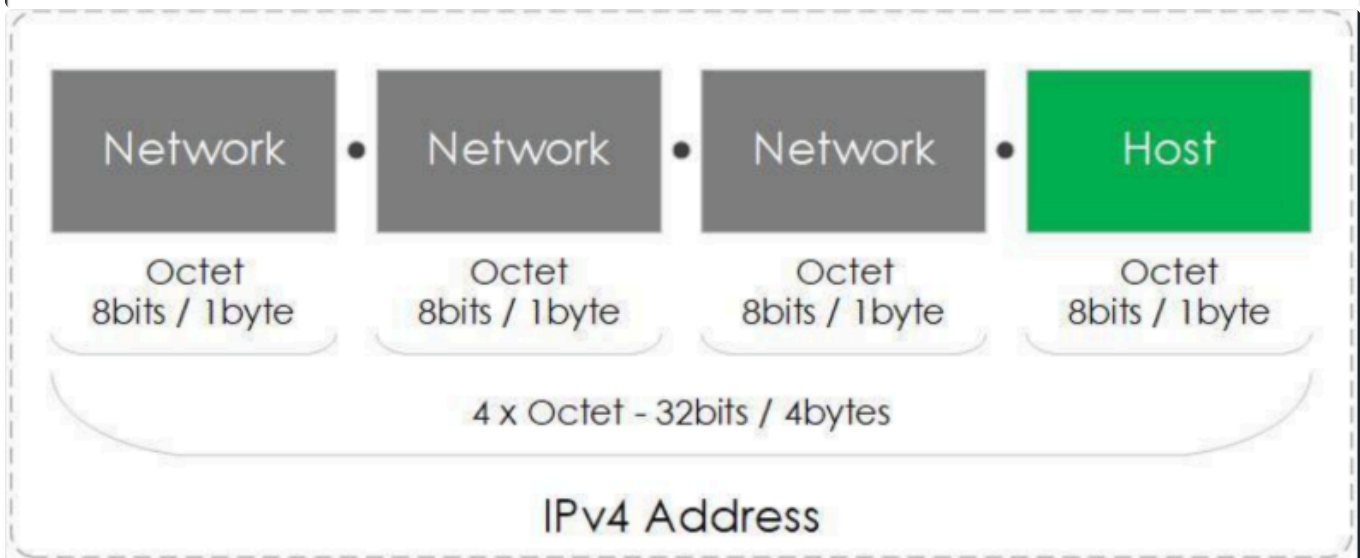
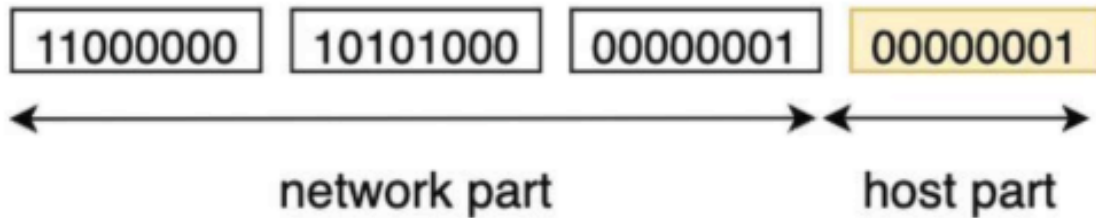
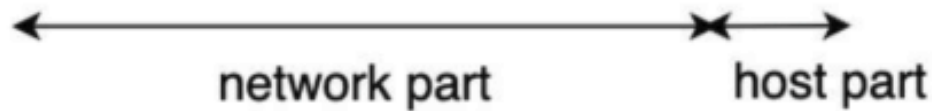
Class 15 IP, Subnet, & MAC Addresses

IP Addressing

An **IP (Internet Protocol) address** is a unique number assigned to every device on a network. It identifies the device and shows its location so data can be sent to the correct destination.

- **Purpose:** Identifies and locates devices on a network.
- **Example:** 192.168.1.10
- **IPv4:** Uses **32 bits**, divided into **4 octets** (8 bits each).
- **Each octet:** Ranges from **0 to 255**.
- **Binary:** Computers store IP addresses in binary.
 - Example:
 - 192 = 11000000
 - 168 = 10101000
- **Why learn binary?** While SOC analysts rarely use binary every day, it is important for understanding **subnetting** and network troubleshooting.

192.168.1.1



Network Part

The **Network Part** of an IP address identifies **which network** a device belongs to. It is like the **city or area** in a postal address.

Example:

- IP Address: 192.168.1.10
- Network Part: 192.168.1 (assuming a /24 subnet mask)

This means all devices with IP addresses like:

- 192.168.1.5

- 192.168.1.20
- 192.168.1.100

belong to the **same network** and can communicate directly without a router. The last part (.10) identifies the specific device on that network.

Host Part

The **Host Part** of an IP address identifies the **specific device** within a network. It is like the **house number** in a city address.

Example:

- IP Address: 192.168.1.10
- Host Part: 10 (assuming a /24 subnet mask)

This means:

- 10 is the unique device number on the 192.168.1 network.
- Every device in the same network must have a **different host number** to avoid IP address conflicts.

Subnetting Basic

What is Subnetting?

Subnetting is the process of dividing a large network into **smaller subnetworks (subnets)**.

Why use subnetting?

- **Better security** – Separates different departments or systems.
- **Better performance** – Reduces network traffic and congestion.
- **Easier management** – Makes networks simpler to organize and troubleshoot.

CIDR Notation

Example: 192.168.1.0/24

- /24 means:

- **24 bits** are used for the **network part**.
- **8 bits** are used for the **host part**.

This creates one network (`192.168.1.0`) with up to **254 usable host IP addresses** (from `192.168.1.1` to `192.168.1.254`).

What is CIDR?

CIDR (Classless Inter-Domain Routing) is a method of assigning IP addresses and creating subnets more **flexibly** than the old class-based system.

Why is CIDR used?

- Efficient use of IP addresses (less waste)
- Flexible subnet creation
- Better network management and routing

What is a /24 Subnet?

A `/24` means:

- **24 bits** are used for the **network part**.
- **8 bits** are used for the **host part**.

Example:

- Network: `192.168.1.0/24`
- Usable host IPs: `192.168.1.1` to `192.168.1.254`
- Total IP addresses: **256**
- Usable host addresses: **254** (excluding the network and broadcast addresses)

In simple terms, a `/24 subnet` divides the IP address so that the first three octets identify the network, and the last octet identifies individual devices on that network.

How Many IP Addresses Are in a /24 Subnet? (Short Explanation)

A `/24 subnet` contains **256 total IP addresses**.

Formula:

- **Total IPs** = $2^{(32 - \text{Subnet Prefix})}$
- $2^{(32 - 24)} = 2^8 = 256$

Reserved IP Addresses:

- **Network Address:** First IP (e.g., 192.168.0.0) – Identifies the subnet.
- **Broadcast Address:** Last IP (e.g., 192.168.0.255) – Used to send data to all devices in the subnet.

Usable IP Addresses:

- $256 - 2 = 254$ usable IPs
- Usable range: 192.168.0.1 to 192.168.0.254

Subnetting Basics

CIDR	Subnet mask	Hosts
/24	255.255.255.0	254
/16	255.255.0.0	65.634
/8	255.0.0.0	16M

Versions of IP Addresses

IPv4 (Internet Protocol Version 4)

- **Address size:** 32-bit
- **Format:** Decimal (e.g., 192.168.1.1)
- **Total addresses:** About **4.3 billion** (2^{32})
- **IPSec:** Optional
- **Status:** Most widely used, but limited addresses have led to address exhaustion.

IPv6 (Internet Protocol Version 6)

- **Address size:** 128-bit
- **Format:** Hexadecimal (e.g., 2001:db8::1)
- **Total addresses:** About **3.4×10^{38}** (2^{128})
- **IPSec:** Built into the protocol suite (support is mandatory in the specification, though use is not always required in practice).

- **Purpose:** Created to overcome IPv4 address shortages and support the growing number of internet-connected devices.

Key Difference

- **IPv4:** 32-bit, decimal format, limited addresses.
- **IPv6:** 128-bit, hexadecimal format, enormous address space, designed for the future Internet.

Classes of IP Addresses

Class A

- **Range:** 1.0.0.0 - 126.255.255.255
- **Network bits:** 8
- **Host bits:** 24
- **Use:** Large networks
- **Example:** Government organizations and large enterprises
- **Note:** 127.x.x.x is reserved for **loopback** (localhost).

Class B

- **Range:** 128.0.0.0 - 191.255.255.255
- **Network bits:** 16
- **Host bits:** 16
- **Use:** Medium-sized networks
- **Example:** Universities and large companies.

Class C

- **Range:** 192.0.0.0 - 223.255.255.255
- **Network bits:** 24
- **Host bits:** 8
- **Use:** Small networks
- **Example:** Home networks and small businesses

Class D

- **Range:** 224.0.0.0 - 239.255.255.255
- **Use:** Multicast communication (sending data to multiple devices at the same time)

- **Example:** Video streaming, online webinars, and live broadcasts

Class E

- **Range:** 240.0.0.0 – 255.255.255.255
- **Use:** Experimental and research purposes
- **Public Use:** Not used for normal public networking or assigning IP addresses to devices.

Summary:

- **Class A:** Large networks, more hosts.
- **Class B:** Medium-sized networks, balanced number of networks and hosts.
- **Class C:** Designed for small networks with fewer hosts.
- **Class D:** Reserved for multicast traffic, not for assigning IP addresses to individual devices.

Types of IP Addresses

1. Public IP Address

- A **Public IP** is assigned by an Internet Service Provider (ISP).
- It is **unique on the Internet** and allows devices to communicate over the internet.
- **Example:** 8.8.8.8

2. Private IP Address

- A **Private IP** is used inside local networks (home, office, etc.).
- It **cannot be accessed directly from the Internet**.
- Common private IP ranges:
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255

3. Static IP Address

- A **Static IP** is a **fixed IP address** that does not change.
- It is commonly used for **servers, websites, and network devices**.
- **Advantage:** Reliable and always reachable.

4. Dynamic IP Address

- A **Dynamic IP** is automatically assigned by a **DHCP server**.
- It can change over time.
- Most home users receive dynamic IP addresses from their ISP.

SOC Perspective

IP Type	Meaning
Private	Internal Traffic
Public	External Connection

Example -

Source: 192.168.1.10

Destination: 45.77.123.66

Internal -> External Communication

MAC Address

What is a MAC Address?

A **MAC (Media Access Control) address** is a **unique hardware identifier** assigned to a network interface (such as a Wi-Fi or Ethernet adapter).

Example: 00:1A:2B:3C:4D:5E

Structure

- **48-bit** address
- Written in **hexadecimal** (six pairs of hexadecimal digits)

Parts of a MAC Address

- **First 24 bits (OUI):** Identifies the **manufacturer** of the device.
- **Last 24 bits (Device ID):** Uniquely identifies the specific network interface.

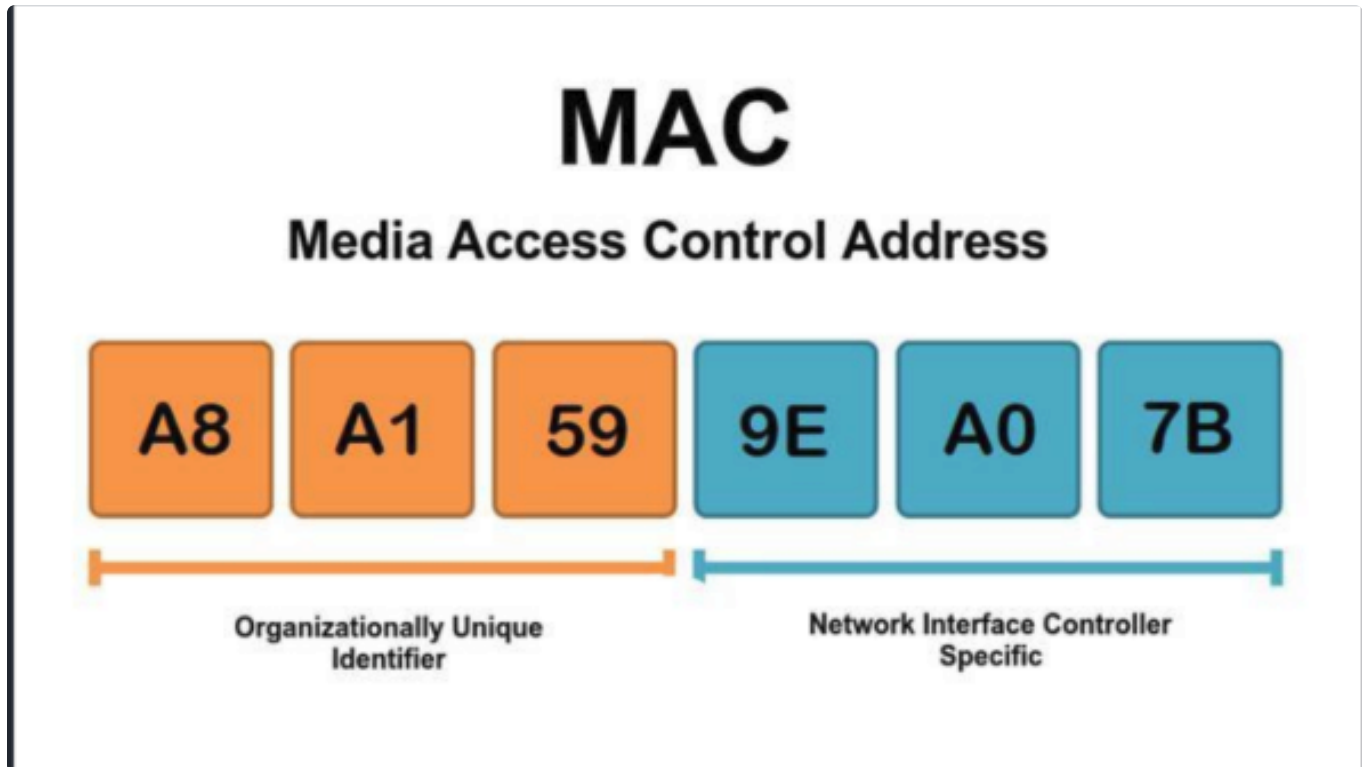
Key Points

- **IP Address:** Identifies a device's **location** on a network and can change.

- **MAC Address:** Identifies the device's **hardware** and is usually permanent.

Easy to remember:

- **IP Address** = Device's network address (can change).
- **MAC Address** = Device's hardware identity (usually fixed).



ARP (Address Resolution Protocol)

ARP (Address Resolution Protocol) is used to map an IP address to a MAC address on a local network.

ARP maps:

- IP Address → MAC Address

Example

A device wants to send data to 192.168.1.1 but only knows its IP address.

It sends an **ARP Request**:

"Who has 192.168.1.1? Tell me your MAC address."

The device with IP 192.168.1.1 replies with its **MAC address**.

Why ARP is Important

- Enables devices on the same LAN to communicate.
- Converts an **IP address** into the **physical (MAC) address** needed to deliver data.

Easy to remember:

- **IP = Logical address**
- **MAC = Physical address**
- **ARP = Finds the MAC address from an IP address**

MAC Address vs IP Address

MAC ADDRESS VERSUS IP ADDRESS

MAC ADDRESS

A unique identifier assigned to a Network Interface Controller (NIC) of the computing device

Stands for Media Access Control Address

Also called physical, hardware or ethernet address

Helps to uniquely identify the device

Assigned by the device manufacturer

Cannot be changed

48 bits (6 bytes) long

Works in the Data Link Layer

IP ADDRESS

A numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication

Stands for Internet Protocol Address

Also called logical address, network or internet address

Helps to identify the connection of a device on the internet

Assigned by an administrator or an ISP

Can be changed

IPv4 - 32 bits (4 bytes)
IPv6 - 128bits (16bytes)

Works in the Network Layer

Real SOC Scenario – ARP Spoofing (Short Explanation)

Scenario:

Multiple devices are showing the **same MAC address** on the network.

Likely Cause:

- **ARP Spoofing (ARP Poisoning)**
- **Man-in-the-Middle (MITM) Attack**

What happens?

An attacker sends **fake ARP replies**, making multiple devices believe that the attacker's MAC address belongs to another device (such as the gateway). As a result, network traffic is redirected through the attacker.

SOC Analyst Action:

- Check the **ARP table** for duplicate or unexpected MAC addresses.
- Compare MAC addresses with known network devices.
- Monitor for suspicious ARP traffic using tools like **Wireshark**.
- Isolate the affected device and investigate further.

Easy to remember:

- **Same MAC on multiple IPs = Possible ARP Spoofing → Potential MITM attack.**